A Novel Adaptive Transmit Diversity Scheme with Simple Feedback

Abstract

A novel adaptive transmit diversity scheme with simple feedback is proposed in this paper for wireless communication systems. The objective of this new diversity technique is to achieve better system performance with less feedback information compared to other transmit diversity methods. With simple linear operations at both the transmitter and receiver, the proposed algorithm requires only 1 bit feedback information for systems with 2 transmit antennas. When there are $M > 2$ antennas at the transmitter, the size of the feedback information required by our transmit diversity scheme is $2M - 2$ bits, which is still significantly smaller than that required by most other close loop transmit diversity techniques. Based on the statistical properties of the received signals, theoretical performance bounds of the proposed transmit diversity bounds are derived. Theoretical analyses and simulation examples show that the proposed transmit diversity scheme outperforms not only the conventional open loop transmit diversity techniques, but also some close loop transmit diversity techniques with more information transmitted in the feedback channel.

I. Introduction

The next generation wireless communication systems are required to provide high quality voice services as well as broadband data services with rates far beyond the limitations of current wireless systems. For example, high speed downlink packet access (HSDPA) [1], which is endorsed by the 3rd generation partnership project (3GPP) standard for WCDMA systems, intends to provide data rates up to 10 Mbps or higher in the downlink channel as opposed to the maximum 384 kbps supported by the enhance data rate for GSM evolution (EDGE) [2], the so-called 2.5G communication standard. To achieve this objective, one of the key technologies employed by emerging wireless systems is antenna diversity, which can effectively combat the adverse effects of multipath fading by providing multiple replicas of the transmitted signal at the receiver. Due to the limited size and cost of the user end device, downlink transmissions favor transmit diversity over receiver diversity, although the latter has been well studied in the literature.

Space time coding, where the data symbols are encoded in both time domain (transmission intervals) and space domain (transmit antenna array), is one of the most common transmit diversity techniques [3]-[6]. Orthogonal space time block code (STBC) was introduced by Alamouti *et.al.* in [3] for systems with exactly two transmit antennas, and full diversity order are achieved by this technique with simple algebraic operations. Space time trellis coding [4] exploits the potential of the multiple antenna arrays by striving to increase both the diversity gains and coding gains of the system, and better performances are achieved at the cost of relative higher encoding and decoding complexity. The techniques mentioned above are designed under the assumption that the transmitter has no knowledge of the fading channels, and they can be classified as open loop transmit diversity.

System performance can be further improved if partial or perfect channel information is available to the transmitter through separate feedback channels (close loop), and this feedback information can be utilized as side information in the design of the transmit diversity algorithm[7]-[14]. In [7] and [8], space time block coding is combined with linear optimum beamforming, and the linear coding matrices are optimized by virtue of the side information of the fading channels. Transmit adaptive array (TXAA), which has been adopted by the 3GPP standard for WCDMA systems, is another close loop transmit diversity system with the transmitted symbols encoded only in the space domain, and optimum performance can be achieved provided the fading channel vector is known perfect to the transmitter[9], [10]. The concept of space encoded transmit diversity is generalized as maximal ratio transmission (MRT) by Lo in [11]. All of the above works require the feedback information to be $M \times N$ complex-valued matrices with elements either directly being the channel impulse response (CIR), or some statistics (*e.g.*, mean or covariance) of the CIR, where M, N are the numbers of transmit antennas and receive antennas, respectively. With the feedback matrices containing $2MN$ real-valued scalars, it's apparent that considerable bandwidth in the reverse link will be consumed by the feedback information. To overcome this problem, suboptimum methods with less feedback information are introduced in [12]-[14]. In [12] and [13], adaptive space time block coding (ASTTD) is proposed with side information being a real-valued vector made up of power ratios of the fading channels, and this side information are used to adjust the power of each transmission antenna. The size of side information is further reduced to one real scalar per transmission interval by Lo in [14]. The suboptimum algorithms presented in [12]-[14] are extensions of STBC, therefore they can

only be used for systems with two transmit antennas [5].

In this paper, a simple binary adaptive transmit diversity scheme is proposed for systems with arbitrary number of transmit antennas, and the objective of this diversity technique is to achieve better system performance with as less feedback information as possible. Unlike the commonly used space time coding scheme, the proposed diversity scheme encodes the data symbols in the space domain, and simple linear operations are employed in both the encoding and decoding process. With the algorithm proposed in this paper, only 1 bit feedback information is required to form the space encoding vector for systems with two transmit antennas, as opposed to complex-valued or real-valued matrices and scalars required by other algorithms. For systems with $M > 2$ transmit antennas, the encoder at the transmitter requires $2M-2$ bits binary feedback information, which is still considerable less than most of the other close loop transmit diversity techniques. For the purpose of analysis and comparisons, theoretical error probability bounds are derived for the proposed transmit diversity technique. Theoretical analyses and simulation results show that our binary adaptive transmit diversity technique outperforms not only the conventional open loop diversity techniques, but also some of the close loop techniques with more information transmitted in the feedback channel.

The rest of this paper is organized as follows. Section II presents the system model. In Section III, the binary adaptive transmit diversity algorithm is first introduced for systems with two transmit antennas, then a modified algorithm is presented when there are more than two transmit antennas present at the transmitter. Section IV derives the theoretical performance bounds of the error probability of the proposed algorithms. Simulation examples are given in Section V, and Section VI concludes the paper.

II. System Model

We consider a system with M transmit antennas and one receive antenna, and the block diagram of the baseband representation of the diversity system is shown in Fig. 1.

At the time instant k, the modulated symbol $s_k \in \mathcal{S}$ is linearly pre-encoded in the space domain by the space encoding vector $\mathbf{p}_k = [p_1(k), p_2(k), \cdots, p_M(k)] \in \mathbb{C}^{1\times M}$ at the transmitter, where $\mathcal{S}$ is the modulation symbol set. The encoded transmit vector is $\mathbf{x}_k = [x_1(k), x_2(k), \cdots, x_M(k)] = \mathbf{p}_k \cdot s_k$, with $x_m(k)$ being transmitted at the mth transmit antenna. In the adaptive transmit diversity described in this

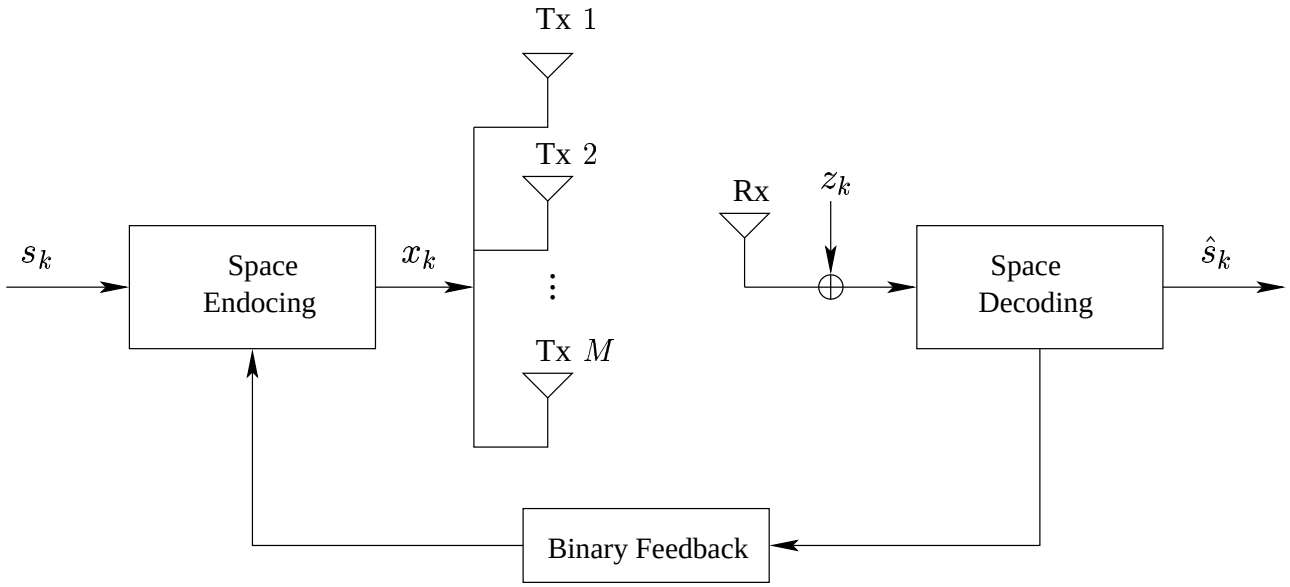


Fig. 1. The block diagram of the system with transmit diversity.

paper, the space encoding vector $\mathbf{p}_k$ is determined at the transmitter by the binary feedback information received from the receiver.

At the receiver, the received signals are the sum of the prorogation signals from all the transmit antennas plus additive noise, and the samples at the receiver can be expressed by

$$\begin{aligned} r(k) &= \sqrt{\frac{E_s}{M}} \cdot \mathbf{x}_k \mathbf{h}_k + z_k, \\ &= \sqrt{\frac{E_s}{M}} \cdot (\mathbf{p}_k \mathbf{h}_k) \cdot s_k + z_k, \end{aligned} \tag{1}$$

where E_s is the sum of the transmit energy of all the transmit antennas, M is the number of transmit antennas, z_k is the additive white Gaussian noise (AWGN) with variance $N_0/2$ per dimension, $\mathbf{h}_k = [h_1(k), h_2(k), \cdots, h_M(k)]^T \in \mathbb{C}^{M\times 1}$ is the time-varying channel impulse response (CIR) of the fading channels, with $h_m(k)$ being the CIR for the fading channel between the mth transmit antenna and the receive antenna, and $(\cdot)^T$ denotes matrix transpose.

With the system model defined in (1), it is well known that the optimum precoding vector $\hat{\mathbf{p}}_k$ that maximizes the output SNR is [9], [10]

$$\hat{\mathbf{p}}_k = \frac{\mathbf{h}_k^H}{\mathbf{h}_k \mathbf{h}_k^H}, \tag{2}$$

where $(\cdot)^H$ denotes matrix Hermitian, and this scheme is called TXAA. However, forming this optimum precoding vector requires the perfect knowledge of the channel CIR vector $\mathbf{h}_k$, which contains M real-valued scalars. Moreover, it will be shown in this paper that the TXAA diversity scheme is very sensitive to quantization error. Hence it is impractical to implement the TXAA scheme in real systems where only very limited resources are allocated to the feedback channel. To balance the system performance and the size of feedback information, a suboptimum adaptive transmit diversity scheme with only binary feedback information is discussed in the next section.

III. Binary Adaptive Transmission Diversity

The binary adaptive transmit diversity scheme is described in this section. The algorithm is first introduced for the simplest system with two transmit antennas, and only one bit feedback information is required to form the precoding vector for such systems. Then a modified algorithm is proposed for systems with $M > 2$ transmit antennas to take into accounts the interactions between all the possible antenna pairs, and $2(M-1)$ bits feedback information is required for the formulation of the precoding vector.

A. Systems with Two Transmit Antennas

The basic idea of the binary adaptive transmission diversity is presented here for the simplest case: systems with two transmit antennas and one receive antenna. In such systems, only one bit feedback information is required by our algorithm for the pre-encoding. The extension of this transmit diversity to more general systems is discussed in the next subsection.

For systems with two transmit antennas, we define the space encoding vector as

$$\mathbf{p}_k = [1, (-1)^{b_k}], \tag{3}$$

where $b_k \in \{0, 1\}$ is the binary feedback information sent out from the receiver. The feedback bit b_k is computed at the receiver based on the estimated CIR $\mathbf{h}_k$ as follows,

$$b_k = \begin{cases} 0, & \text{if } \Re\{h_1(k)h_2^*(k)\} > 0, \\ 1, & \text{otherwise}, \end{cases} \tag{4}$$

where $h_m(k)$ is the time-varying channel impulse response (CIR), $(\cdot)$ denotes complex conjugate, and the operation $\Re(\cdot)$ returns the real part of the operand. It is assumed here that the receiver has perfect knowledge of the fading channels, and there is no error in the feedback channel.

With the definition of the space encoding vector $\mathbf{p}_k$ in (3), the transmitted signal vector is $\mathbf{x}_k = [s_k, (-1)^{b_k} s_k]$. Replacing the vector $\mathbf{x}_k$ in (1), we have the received sample as

$$r(k) = \sqrt{\frac{E_s}{2}}[h_1(k) + h_2(k)(-1)^{b_k}] \cdot s_k + z_k. \tag{5}$$

In receivers with coherent detection, the received sample $r(k)$ is multiplied by $(\mathbf{p}_k \mathbf{h}_k)^H = h_1^*(k) + h_2^*(k)(-1)^{b_k}$ to form the decision variable $y(k)$,

$$\begin{aligned} y(k) &= (h_1^*(k) + (-1)^{b_k} h_2^*(k)) \cdot r_k, \\ &= \sqrt{\frac{E_s}{2}} \left[|h_1(k)|^2 + |h_2(k)|^2 + (-1)^{b_k} \cdot 2\Re\{h_1(k)h_2^*(k)\}\right] s_k + v_k, \end{aligned} \tag{6}$$

where $v_k = \left[h_1^*(k) + (-1)^{b_k} h_2^*(k)\right] \cdot z_k$ is the noise component of the decision variable. It can be easily shown that the variance of noise component v_k is

$$\sigma_v^2 = \left[|h_1(k)|^2 + |h_2(k)|^2 + (-1)^{b_k} \cdot 2\Re\{h_1(k)h_2^*(k)\}\right] \cdot N_0. \tag{7}$$

It can be seen from (4) that

$$(-1)^{b_k} \cdot 2\Re\{h_1(k)h_2^*(k)\} = 2|\Re\{h_1(k)h_2^*(k)\}|, \tag{8}$$

thus the instaneous output SNR γ at the receiver can be written by

$$\gamma = \frac{\gamma_0}{2} \cdot \left[|h_1(k)|^2 + |h_2(k)|^2 + 2|\Re\{h_1(k)h_2^*(k)\}|\right], \tag{9a}$$

$$= \gamma_0 \cdot (g_c + g_b), \tag{9b}$$

where $\gamma_0 = \frac{E_s}{N_0}$ is the SNR without diversity, and the diversity gains g_c and g_b are defined as

$$g_c = \frac{1}{2}[|h_1(k)|^2 + |h_2(k)|^2], \tag{10a}$$

$$g_b = |\Re\{h_1(k)h_2^*(k)\}|^2. \tag{10b}$$

g_c is the conventional diversity gain, which is the same as the diversity gain of the orthogonal space time block coding (STBC) [3], while g_b is the extra diversity gain contributed by the binary feedback information. From (10) we can see that, with only 1 bit feedback information, the output SNR of

the proposed transmission diversity scheme is always better than the orthogonal STBC, although the transmitted signals in the proposed scheme are only encoded in the space domain.

The algorithm described in this sub-section are for systems with two transmit antennas. when there are more than 2 antennas at the transmitter, the interactions between all the sub-channels will make it difficult to keep higher diversity gains with only 1 bit feedback information. Therefore a modified transmit diversity scheme with $2M-2$ bits feedback information is presented next for systems with $M > 2$ transmit antennas.

B. Systems with More Than Two Transmit Antennas

For systems with $M > 2$ transmit antennas, we define the space encoding vector as

$$\mathbf{p}_k = \left[\begin{array}{cccc} 1 & \exp[\frac{i \cdot q_2(k)\pi}{2}] & \cdots & \exp[\frac{i \cdot q_M(k)\pi}{2}] \end{array} \right], \tag{11}$$

where $i^2 = -1$, and $q_m(k) \in \{0, 1, 2, 3\}$, for $m = 2, 3, \cdots, M$, are the feedback information from the receiver. For consistence of representation, we let $q_1(k) = 0$, for $\forall k$. By such definitions, each $q_m(k)$ contains two bits of information, and there are totally $2(M-1)$ bits feedback information used to form the space encoding vector $\mathbf{p}_k$.

Combining (1) and (11), we can write the received sample $r(k)$ as

$$r(k) = \sqrt{\frac{E_s}{M}} \left\{ \sum_{m=1}^{M} \exp\left[\frac{i \cdot q_m(k)\pi}{2}\right] h_m(k) \right\} \cdot s_k + z_k. \tag{12}$$

At the decoder, the decision variable $y(k)$ is obtained by multiplying the received sample $r(k)$ with $(\mathbf{p}_k \mathbf{h}_k)^H$, and it can be written by

$$\begin{aligned} y(k) &= \sqrt{\frac{E_s}{M}} \left| \sum_{m=1}^{M} \exp[\frac{i \cdot q_m(k)\pi}{2}] h_m(k) \right|^2 \cdot s_k + v_k, \\ &= \sqrt{\frac{E_s}{M}} (g_c + g_b) \cdot s_k + v_k, \end{aligned} \tag{13}$$

where $v_k = (\mathbf{h}_k^T \mathbf{p}_k)^* \cdot z_k$ is the noise component with variance $|\mathbf{p}_k \mathbf{h}_k|^2 \cdot N_0$, and the diversity gains g_c and g_b are defined as

$$g_c = \frac{1}{M} \sum_{m=1}^{M} |h_m(k)|^2, \tag{14a}$$

$$g_b = \frac{2}{M} \sum_{m=1}^{M} \sum_{n=m+1}^{M} \Re \left\{ h_m(k) h_n^*(k) \exp\left[i \cdot \pi \frac{q_m(k) - q_n(k)}{2} \right] \right\}. \tag{14b}$$

In the equations above, the conventional diversity gain g_c is fixed for a certain value of M, while the feedback diversity gain can be maximized by appropriately choosing the feedback information $q_m(k)$.

With the $2M-2$ bits of information, we can maximize g_b by selecting $q_m(k)$ such that all the summed elements of g_b are non-negative. One of the summed element of g_b can be written by

$$\Re\left\{h_m(k)h_n^*(k)\exp\left[i\cdot\pi\frac{q_m(k)-q_n(k)}{2}\right]\right\}=|h_m(k)||h_n(k)|\cos(\Delta\theta_{mn}), \tag{15}$$

where $\Delta\theta_{mn}=\theta_m-\theta_n+\frac{q_m(k)-q_n(k)}{2}\pi$ with $\theta_m\in[0,2\pi)$ being the phase of $h_m(k)$. The terms described in (15) will be non-negative if the following condition is satisfied

$$|\Delta\theta_{mn}|\le\pi/2, \qquad \text{for } \forall m\neq n. \tag{16}$$

One direct way to satisfy this maximization condition described in (16) is adjusting $q_m(k)$ so that all the rotated phases $\theta_m+\frac{q_m(k)}{2}\pi$, for $m=1,2,\cdots,M$, are in the same coordinate quadrant. Without loss of generality, we keeps the phase θ_1 of the first sub-channel $h_1(k)$ unchanged, and rotates the phases θ_m of all the rest sub-channels by $\frac{q_m(k)}{2}\pi$ so that the rotated phases are in the same quadrant as θ_1. It will be shown next that only 2 bits information are required to form each $q_m(k)$ to achieve this goal.

If we label the four quadrants of the Cartesian coordinate system as shown in Fig. 2, then the quadrant number of any angle $\phi\in[0,2\pi)$ can be computed as $\lceil\frac{2\phi}{\pi}\rceil$, where $\lceil\cdot\rceil$ denotes rounding to the nearest bigger integer. With the analyses above, the feedback information $q_m(k)$ for $m=2,3,\cdots,M$ can be computed at the receiver based on the phases of the estimated CIRs

$$q_m(k)=\left\lceil\frac{2\theta_1}{\pi}\right\rceil-\left\lceil\frac{2\theta_m}{\pi}\right\rceil. \tag{17}$$

An example is shown in Fig. 2 with θ_1 in quadrant II and θ_m in quadrant IV. With (17), we can get that $q_m(k)=-2$, which corresponds to rotate θ_m by π radians clockwise, and the rotated phase $\theta_m-\frac{q_m(k)}{2}\pi$ is in quadrant II.

The feedback information defined in (17) guarantees that all the elements described in (15) are non-negative for $\forall m\neq n$, and the maximized diversity gain g_b contributed by the feedback information can be written by

$$g_b=\frac{2}{M}\sum_{m=1}^{M}\sum_{n=m+1}^{M}|h_m(k)||h_n(k)||\cos(\Delta\theta_{mn})|. \tag{18}$$

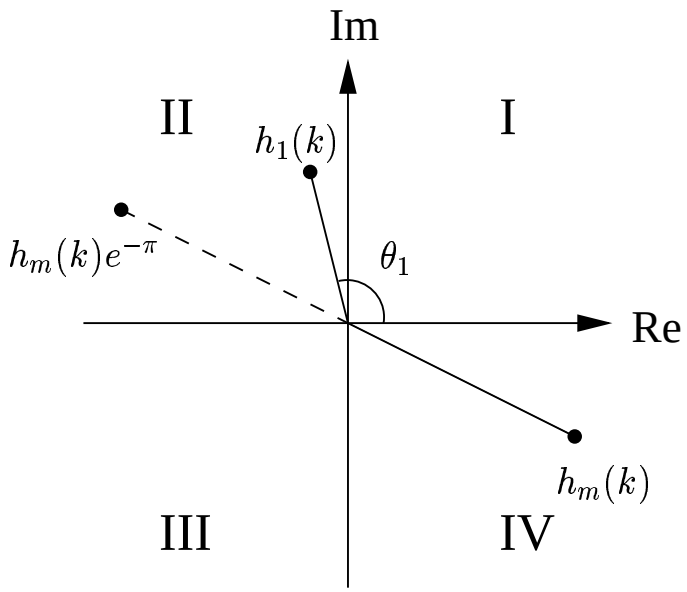


Fig. 2. The four quandrants of the coordinate system.

Combining (13), (14) and (18), we will have the output SNR γ at the detector

$$\gamma = \gamma_0 \cdot \left[\frac{1}{M} \sum_{m=1}^{M} |h_m(k)|^2 + \frac{2}{M} \sum_{m=1}^{M} \sum_{n=m+1}^{M} |h_m(k)||h_n(k)| \cos(\Delta\theta_{mn}) | \right], \tag{19a}$$

$$= \gamma_0 \cdot (g_c + g_b), \tag{19b}$$

where $\gamma_0 = \frac{E_s}{N_0}$ is the SNR without diversity, and the diversity gains g_c and g_b are given in (14a) and (18), respectively.

Before concluding this section, we have the following two remarks about our transmit diversity scheme.

Remark 1: The transmit diversity algorithm proposed for systems with more than two transmit antennas can also be implemented for systems with two transmit antennas, and the required number of feedback bits is $2 \times (2 - 1) = 2$, which is one bit more than that required by the transmit diversity described in Section III-A. This extra bit of feedback information will contribute to slight performance improvement. two transmit antennas.

Remark 2: The conventional STBC and close loop technique based on the orthogonal STBC [12]-[14] can only be implemented for systems with exactly two transmit antennas, while the transmit diversity schemes presented in this paper can be used for systems with arbitrary number of transmit antennas. This is extremely useful for the high speed downlink data transmission of next generation wireless communication

systems, where higher diversity order are required to guarantee high data throughput in the downlink with multiple transmit antennas and one receive antenna.

IV. Performance Analysis

In this section, theoretical performances of the proposed transmit diversity scheme in uncoded communication systems with binary phase shift keying (BPSK) modulation in Rayleigh fading channels are analyzed. Since it is difficult to get the exact expressions of the error probabilities of the proposed algorithms, we resort to the upper bound and lower bound of the error probabilities. These theoretical performance bounds for uncoded communication systems will give us some insights of the performance of the proposed algorithms, and they can be used as guides for the design of real communication systems.

The error probability of a system over fading channels can be computed by

$$P(E) = \int_0^{+\infty} P(E|\gamma) d\gamma, \tag{20}$$

where $P(E|\gamma)$ is the conditional error probability (CEP), and γ is the instantaneous output SNR. For systems with BPSK modulation, the CEP can be written by [15, eqn. (23)]

$$P(E|\gamma) = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} \exp\left[-\frac{\gamma}{\sin^2\theta}\right] d\theta. \tag{21}$$

To evaluate the error probability $P(E)$, we need to know the probability density function (pdf) or characteristic function (CHF) of the output SNR γ. However, with the expressions of γ given in (9) for systems with two transmit antennas and in (19) for systems with more than two transmit antennas, it is difficult to get the closed form expressions of the pdf or CHF of γ. Therefore, we resort to the upper and lower bound of the error probability instead of directly evaluating the exact error probability.

The upper bound of the error probability can be obtained by setting the feedback diversity gain $g_2 = 0$, and the corresponding SNR γ^U becomes

$$\gamma^U = \frac{\gamma_0}{M} \sum_{m=1}^{M} |h_{k_m}|^2. \tag{22}$$

The SNR γ^U scaled by the factor M has similar form as the output SNR of maximal ratio combiner (MRC). For Rayleigh fading channel, γ^U is χ^2 distributed with $2M$-degree of freedom, and the upper bound error probability $P^U(E)$ is [15, eqn. (23)]

$$P^U(E) = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} \prod_{m=1}^{M} \left(1 + \frac{\bar{\gamma}_m}{\sin^2\theta}\right)^{-1} d\theta, \tag{23}$$

where $\bar{\gamma}_m$ is the average SNR of the mth sub-channel given by

$$\bar{\gamma}_m = \frac{\gamma_0}{M} \cdot E(|h_m(k)|^2) = \frac{\gamma_0}{M} \cdot \sigma_m^2, \tag{24}$$

with σ_m^2 being the variance of the CIR $h_m(k)$.

Next we are going to find the lower bound of the system error probability $P^L(E)$ with the help of the output SNR. For systems with two transmit antennas, the output SNR is given in (9), from which we can see that the feedback diversity gain g_b satisfies

$$g_b = |\Re\{h_1(k)h_2^*(k)\}| \leq |h_1(k)||h_2(k)|, \tag{25}$$

and this inequality is based on the fact that, for any complex number a, we always have $|\Re(a)| \leq \sqrt{\Re(a)^2 + \Im(a)^2} = |a|$, where $\Im(\cdot)$ denotes the imaginary part of the operand, and the equality holds when $\Im(a) = 0$. The feedback diversity gain of the transmit diversity scheme for systems with more than two transmit antennas is given in (18), and it can be easily seen that

$$g_b \leq \frac{2}{M} \cdot \sum_{m=1}^{M} \sum_{n=m+1}^{M} |h_m(k)||h_n(k)|. \tag{26}$$

The two inequalities of (25) and (26) can be written under one equation as of (26), and we define the SNR corresponding to lower bound error probability as

$$\begin{aligned} \gamma^L &= \frac{\gamma_0}{M} \cdot \left[\sum_{m=1}^{M} |h_m(k)|^2 + 2 \cdot \sum_{m=1}^{M} \sum_{n=m+1}^{M} |h_m(k)||h_n(k)| \right], \\ &= \frac{\gamma_0}{M} \cdot \left[\sum_{m=1}^{M} |h_m(k)| \right]^2. \quad \text{for } M = 2, 3, \cdots. \end{aligned} \tag{27}$$

It is interesting to note that the SNR γ^L has exactly the same form as the output SNR of coherent equal gain combiner (EGC) [16, eqn. (1)] with M receive antennas and one transmit antenna. Therefore, the lower bound error probability $P^L(E)$ can be expressed in the same form as the error probability expression for EGC receivers [16, eqn. (20)]

$$P^L(E) = \frac{1}{2} - \frac{1}{\pi} \int_0^{\pi/2} \frac{\exp(-\tan\theta)}{\sin 2\theta} \Im\left\{ \phi(2\sqrt{\tan\theta}) \right\} d\theta, \tag{28}$$

where the function $\phi(x)$ is defined as

$$\phi(x) = \prod_{m=1}^{M} \left[{}_1F_1\left(1, \frac{1}{2}; -\frac{\bar{\gamma}_m x^2}{4}\right) + i \cdot x \sqrt{\frac{\pi \bar{\gamma}_m}{4}} \exp\left(-\frac{\bar{\gamma}_m x^2}{4}\right) \right], \tag{29}$$

and ${}_1F_1(a,b;c)$ is the confluent hypergeometric function of the first kind, and $\bar{\gamma}_m$ is the average SNR of the mth sub-channel as defined in (24).

With the upper bound and lower bound error probability given in (23) and (28), the error probability $P(E)$ of the proposed algorithms in uncoded communication systems can be expressed by

$$P^{L}(E) \leq P(E) \leq P^{U}(E). \tag{30}$$

One interesting observation of the upper bound and lower bound error probability of the proposed diversity scheme is that the systems with SNR analog to EGC receiver outperforms systems with SNR similar to MRC receiver. However, for diversity receivers, it is well known that the performance of the MRC receiver is always better than that of the EGC receiver. This seemingly dilemma can be accounted by the fact that in transmit diversity, the transmit power E_s is distributed among all the transmit antennas. The SNR γ_{U} corresponds to $P^{U}(E)$ is the SNR of MRC receiver divided by the number of transmit antennas, $\gamma^{U} = \frac{\gamma_{MRC}}{M}$. However, for EGC receivers, the AWGN of the M receive antennas are summed together directly, which results in a noise variance of $M \cdot N_0$. The up scaling of the noise variance in EGC receiver is counterfeited by the down scaling of the transmit power in transmit diversity. Therefore γ_{L} of transmit diversity is exactly the same as the SNR of EGC receivers, $\gamma^{L} = \gamma_{EGC}$. Although $\gamma_{EGC} \leq \gamma_{MRC}$, it can be easily proven that

$$\gamma^{L} = \gamma_{EGC} = \frac{\gamma_0}{M}\left[\sum_{m=1}^{M}|h_m(k)|\right]^2 \geq \frac{\gamma_0}{M}\sum_{m=1}^{M}|h_m(k)|^2 = \frac{\gamma_{MRC}}{M} = \gamma^{U}. \tag{31}$$

The error probabilities given in this section provide the performance bounds for our transmit diversity schemes, and they can be used as references for the analyses of the proposed method.

V. Simulation Examples

Simulations examples for both uncoded systems and 3GPP HSDPA systems are given in this section to evaluate the performance of the binary adaptive transmit diversity scheme proposed in this paper.

Let's first investigate the bit error rate (BER) performance of the transmit diversity schemes in uncoded wireless systems experiencing Rayleigh flat fading. In the simulations, the symbol rate of the uncoded systems is 240 ksym/sec, and the maximum Doppler frequency is 10 Hz. It is assumed that the receiver has perfect knowledge of the fading channels.

In Fig. 3 are shown the uncoded BER performances of various transmit diversity schemes for BPSK systems with 2 transmit antennas and 1 receive antenna. It is assumed that there is no feedback delay in this example. As discussed in Section III, both the algorithms presented in Section III-A and III-B can be used for systems with 2 transmit antennas, but with different number of feedback bits. As expected, the performance of the system with 2-bit feedback information is approximately 0.4 dB better than the system with only 1 bit feedback information, and the performance difference is mainly contributed by the extra bit of feedback information. However, it can be seen from Fig. 3 that both of the two proposed algorithms outperform the standard orthogonal STBC, and the performance improvement can be up to 2 dB. Moreover, we can see that the theoretical lower bound presented in (28) is only 0.2 dB away from the simulation results, which means the lower bound is very tight.

Since the proposed algorithms are close loop transmit diversity, for comparison purposes, in Fig. 3 we also show the BER performances of two other close loop transmit diversities: TXAA with quantized feedback and ASTTD with real-valued vector feedback. From the figure we can see that TXAA is very sensitive to the quantization error. If there are only 2-bit quantized feedback (1 bit quantization for each sub-channel), the performance of TXAA is even worse than the open loop STTD; if the quantized feedback is increased to 4 bits (2 bits quantization for each sub-channel), the performance of TXAA is still a little bit worse than the proposed scheme with only 1 bit feedback. Moreover, the performance of the proposed algorithm with 1 bit feedback is even better than ASTTD with the feedback being 2×1 real-valued vector. This example demonstrates that our adaptive transmit diversity schemes outperforms not only conventional open loop transmit diversity techniques, but also close loop algorithms which require more information in the feedback channel than our algorithms.

STTD and ASTTD can be implemented for systems with at most 2 transmit antennas. However, our transmit diversity algorithm can be used for systems with arbitrary number of transmit antennas. Fig. 4 and 5 show the BER performances of the transmit diversity systems with different number of transmit antennas. For the purpose of comparison, the BER performances of TXAA with $2M$ bits quantized feedback information (2 bits quantization for each sub-channel) are also illustrated in the figures. From these two figures, we have the following observations. 1) The performance of the proposed algorithm improves almost linearly with the increasing number of transmit antennas; 2) the proposed algorithm with

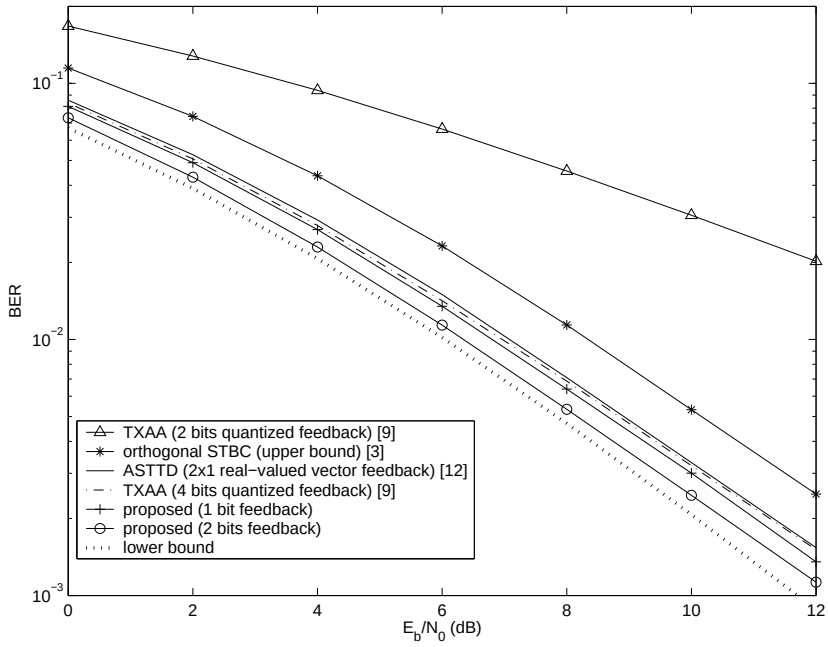


Fig. 3. Comparison of various transmit diversity schemes for systems with 2 transmit antennas

$2M-2$ bits feedback always outperforms the quantized TXAA with $2M$ bits feedback; 3) the performance improvement of the theoretical upperbound tend to zero when $M > 4$; 4) the theoretical lowerbound is very close to the empirical simulation results of the proposed algorithm.

It is worth pointing out that the theoretical upper bound corresponds to the performance of orthogonal STBC. Even real orthogonal design exists for $M > 2$ (*e.g.*, $M = 4$, $M = 8$, [5]), it can be seen from Fig. 5 that nearly no extra performance gain can be achieved with orthogonal STBC when $M > 4$. However, with the algorithm proposed in this paper, the system performance improves almost linearly with the number of the transmit antennas.

In all of the examples above, the feedback delay is assumed to be 0. However, there is always feedback delay in real communication systems. Now we are going to analyze the effects of feedback delay on the performance of the proposed algorithms. Fig. 6 shows the system performances versus feedback delay. From Fig. 6 we can see that, for systems with various number of transmit antennas and modulation schemes, there is no apparent performance degradation with the increase of the feedback delay up to 600 symbol period (2.5 ms).

We have shown the performances of the proposed transmit diversity in uncoded TDMA systems. The next example is used to illustrate the performance of the algorithms in HSDPA WCDMA systems [1].

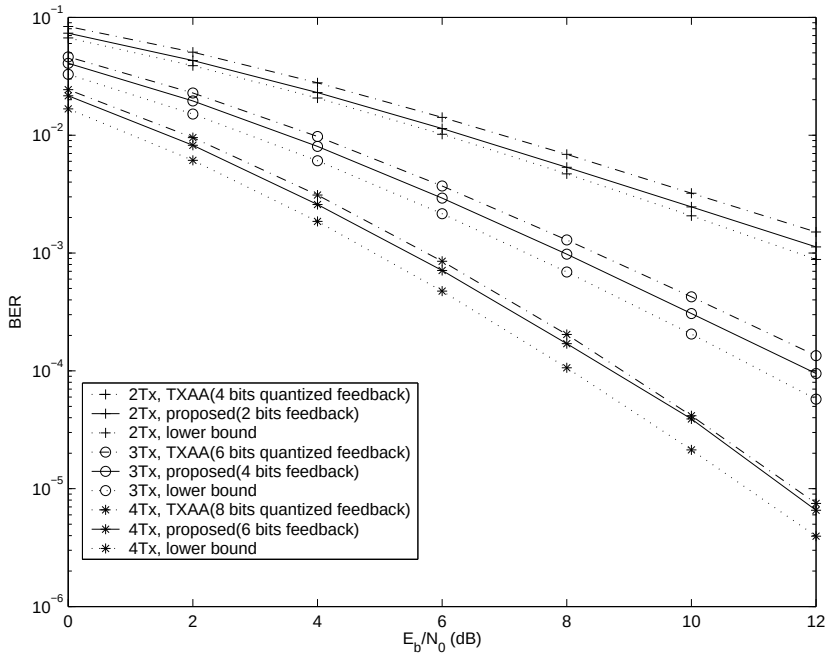


Fig. 4. Comparison the proposed algorithm with quantized TXAA for systems with different number of transmit antennas

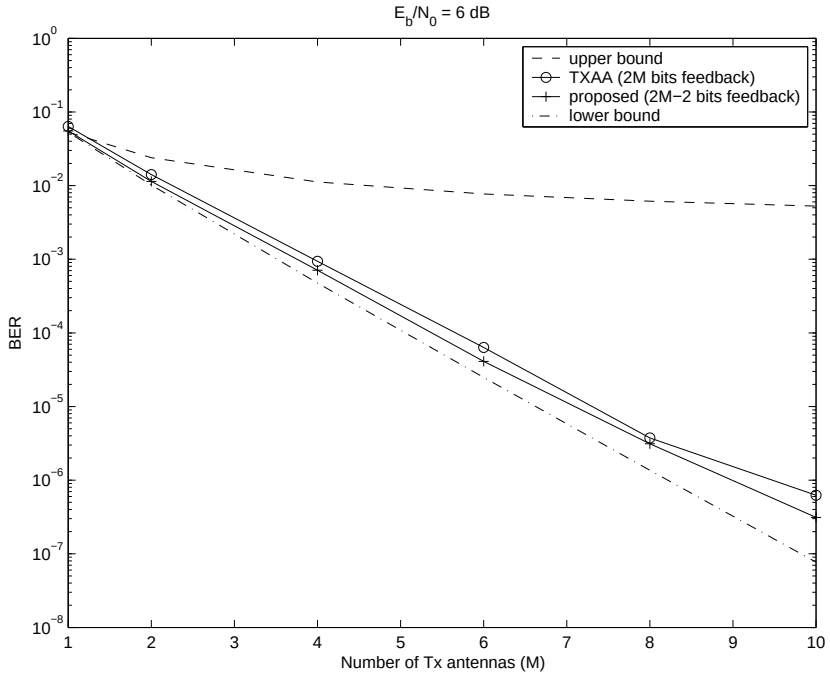


Fig. 5. The effects of transmit antenna number on the performance of the proposed algorithm.

The simulation parameters are defined according to the HSDPA technical specifications [1] and are shown in Table 1.

In Table 1, I_{or} is the carrier power, I_{oc} is the power of interference and additive noise, E_c is the

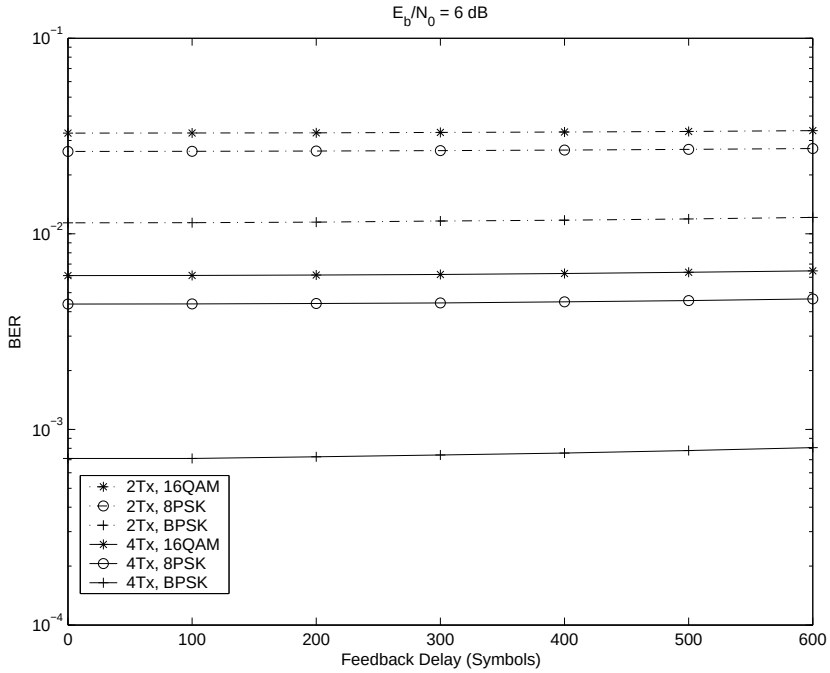


Fig. 6. The effects of the feedback delay on the performance of the proposed algorithm.

power of the signals, CPICH is the acronym for common pilot channel, TTI stands for transmission time interval, and every frame contains 3 TTIs. With the parameters defined in Table 1, we can compute the data transmission rate R as

$$R = R_c \cdot \frac{N_c}{\text{SF}} \cdot R_t \cdot \log_2 64 = 10.8\text{Mbps} \tag{32}$$

The frame error rate (FER) of different transmit diversity algorithms in HSDPA systems are shown in Fig. 7, and the results are consistent with those obtained from uncoded TDMA systems. This example justifies the implementation of the newly proposed algorithms in practical communication systems.

VI. Conclusions

In this paper, a novel binary adaptive transmit diversity scheme is proposed for wireless communication systems for the purpose of achieving better systems performance with less feedback information compared to other transmit diversity methods. With the algorithm proposed in this paper, only 1 bit feedback information is required for systems with two transmit antennas, and for systems with $M > 2$ transmit antennas, the required size of feedback information is $2M - 2$ bits, which is still significantly smaller than required by most other close loop transmit diversity techniques. Simulation results show that the

performance of the proposed algorithms is up to 2 dB better than the performance of open loop STBC. When compared with other close loop techniques, the performance is better than or nearly the same as the performance of some close loop algorithms, even much more feedback information is required by these close loop algorithms than our approach.

Table 1. Simulation Parameters for 3GPP HSDPA Systems

Carrier Frequency	2 GHz
Spreading Factor (SF)	16
Number of Multicodes (N_c)	10
Frame Length	2ms
Chip Rate (R_c)	3.84 Mbps
CPICH power	$10\% I_{or}$
E_c/I_{or}	70%
I_{or}/I_{oc}	variable
Channel Coding	Turbo, rate $R_t = 3/4$
Fading Model	one path Rayleigh
Correlation Model	i.i.d.
Channel Estimation	perfect
Modulation	64 QAM
Feedback Delay	4 TTI

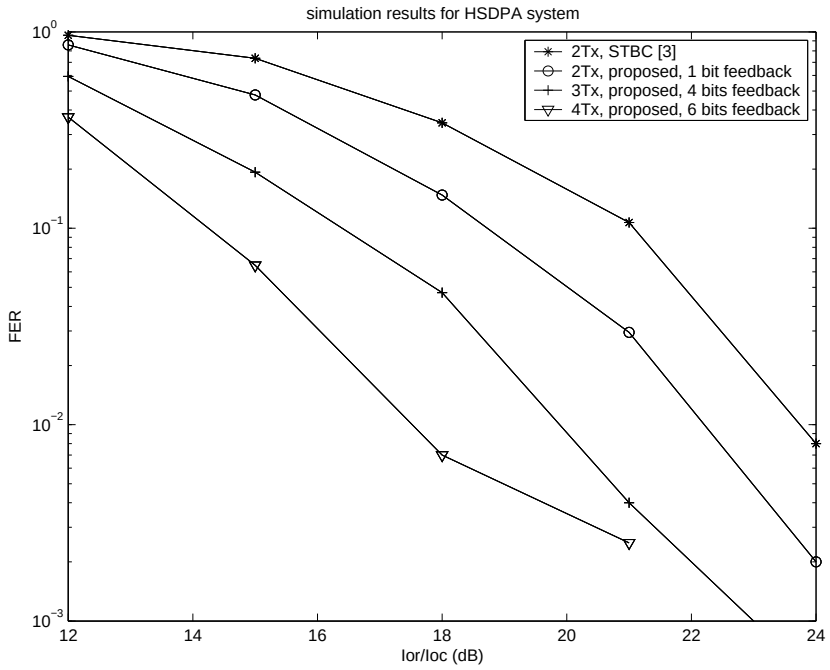


Fig. 7. Performances of transmit diversity in turbo coded HSDPA system.

REFERENCES

[1] 3GPP: 3GPP TR25.848 v4.0.0, "3GPP technical report: Physical layer aspects of utra high speed downlink packet access,", Mar. 2001. (available at http://www.3gpp.org)

[2] ETSI. GSM 05.05, "Radio transmission and reception," ETSI EN 300 910 V8.5.1, Nov. 2000.

[3] S.M. Alamouti, "A simple transmit diversity technique for wireless communications," *IEEE J. Select. Area Commun.*, vol.16, pp.1451-1458, Oct. 1998.

[4] V. Tarokh, N. Seshadri, and A.R. Calderbank, "Space-time codes for high data rate wireless communication: performance criterion and code construction," *IEEE Trans. Info. Theory*, vol.44, pp.744-765, Mar. 1998.

[5] V. Tarokh, H. Jafarkhani, and A.R. Calderbank, "Space-time block codes from orthogonal designs," *IEEE Trans. Info. Theory*, vol.45, pp.1456-1467, Jul. 1999.

[6] Y. Xin, Z. Wang, and G.B. Giannakis, "Space-time diversity systems based on linear constellation precoding," *IEEE Trans. Wireless Commun.*, vol.2, pp.294-309, Mar. 2003.

[7] G. Jongren, M. Skoglund, and B. Ottersten, "Combining beamforming and orthogonal space-time block coding," *IEEE Trans. Info. Theory*, vol.48, pp.611-627, Mar. 2002.

[8] S. Zhou, G.B. Giannakis, "Optimal transmitter eigen-beamforming and space-time block coding based on channel mean feedback," *IEEE Trans. Signal Processing*, vol.50, pp.2599-2613, Oct. 2002.

[9] K. Rohani, M. Harrison, and K. Kuchi, "A comparison of base station transmit diversity methods for third generation cellular standards," in *Porc. IEEE Veh. Techno. Conf. VTC'99 Spring*, pp.351-355, May, 1999.

[10] R.T. Derryberry, S.D. Gray, D.M. Ionescu, G. Mandyan, and B. Raghothaman, "Transmit diversity in 3G CDMA systems," *IEEE Commun. Mag.*, vol.40, pp.68-75, Apr. 2002.

[11] T. Lo, "Maximum ratio transmission," *IEEE Trans. Commun.*, vol.47, pp.1458-1461, Oct. 1999.

[12] Huawei:TSGR1-02-0711 "STTD with adaptive transmitted power allocation," May, 2002. (available at http://www.3gpp.org)

[13] J.H. Horng, L. Li, and J. Zhang, "Adaptive space-time transmit diversity for MIMO systems," in *Proc. IEEE Veh. Techno. Conf. VTC'03 Spring*, pp.1070-1073, Apr. 2003.

[14] T. Lo, "Adaptive space-time transmission with side information," *IEEE Trans. Veh. Technol.*, vol.00, pp.0000-0000, 00, 2003.

[15] M.K. Simon, and M.-S. Alouini, "A unified approach to the performance analysis of digital communication over generalized fading channels," *Proc. of IEEE*, vol.86, pp.1860-1877, Sep. 1998.

[16] A. Annamalai, C. Tellambura, and V.K. Bhargava, "Equal-gain diversity receiver performance in wireless channels," *IEEE Trans. Commun.*, vol.48, pp.1732-1745, Oct. 2000.